

Django supports several capture types, with the three most common being the following:

- `str`—Any non-empty string excluding the forward-slash (/) path separator
- `int`—Any integer zero or more
- `slug`—A special string value consisting of letters, numbers, and hyphens

The capture specifier is part of the path-matching process. If your URL specifies an integer component, but your user sends in a non-integer, the path does not match. This means your view doesn't get called. Django treats it as a 404 without you needing to worry about it.

Up until now, you have registered your URL path objects directly in the main `urls.py` file. It is actually good practice to associate the URLs for an app with the app. You can do this by creating a new URL-routing file in the app and then including its contents in the main file.

A URL pattern file in an app uses the exact same structure as the main file. Create `RiffMates/bands/urls.py` like you see in the following listing.

Listing 4.5 App specific URL pattern declaration file for bands

```
# RiffMates/bands/urls.py
from django.urls import path

from bands import views

urlpatterns = [
    path('musician/<int:musician_id>/', views.musician, name="musician"),
]
```

A path object with an integer capture specifier

Using separate URL-routing files for each app tends to simplify the code you need in the file. Without them, the main file needs to alias the view module, as all apps have the same name for their views: `views.py`. Having a bands-app-specific URL file only uses the bands app's `views.py` file, so the alias is unnecessary.

Django does not automatically find and register app specific URL routing files—you need to register them. The `django.urls` module provides a function called `include()` that enables you to include all the URLs in a module. The `include()` function gets used as part of a path object, registering all the URLs in the included module under a path. This way, all the URLs in a module start with the same path prefix. To avoid the need to import many modules, the `include()` function takes a string that names the module whose content is to be included. Open `RiffMates/RiffMates/urls.py`, and make the changes you see in the following listing.

Listing 4.6 Registering all the URL routes in the bands app

```
# RiffMates/RiffMates/urls.py
...
from django.urls import path, include
...
```

Add the `include()` function to the import line.